Introduction

Factorial trials test two or more interventions in a single experiment: participants are randomised to each factor independently, producing 2x2 (or higher) cells. The design is efficient if the interventions act independently (no interaction); otherwise the interaction itself becomes the primary finding.

Prerequisites

ANOVA; interactions.

Theory

In a 2x2 design, four cells: control, A only, B only, A + B. Main effects are estimated by averaging over the other factor; the interaction compares observed A + B effect to the sum of individual effects.

If interaction is negligible, factorial is efficient: same power as two separate trials with half the total sample size.

Assumptions

Treatments do not interact (or the interaction is the inferential target); randomisation is to each factor independently; outcome is measured under identical conditions across cells.

R Implementation

```r
set.seed(2026)
n_per_cell <- 40
A <- factor(rep(c("no", "yes"), each = 2 * n_per_cell))
B <- factor(rep(rep(c("no", "yes"), each = n_per_cell), 2))

# Simulate additive effects, mild positive interaction
y <- rnorm(4 * n_per_cell) +
     ifelse(A == "yes", 0.5, 0) +
     ifelse(B == "yes", 0.3, 0) +
     ifelse(A == "yes" & B == "yes", 0.2, 0)

fit <- lm(y ~ A * B)
summary(fit)$coefficients
anova(fit)
```

Output & Results

Main-effect estimates for A and B plus the interaction term. The interaction is small, consistent with the simulated +0.2.

Interpretation

"The factorial analysis estimated a main effect of A = 0.48, B = 0.31, with a small positive interaction (0.22, p = 0.14). Main-effect analyses are interpretable in the absence of significant interaction."

Practical Tips

- Pre-specify the interaction test and its interpretation; a non-significant test does not guarantee additivity.
- Factorial trials are under-powered for interactions unless specifically designed for them.
- Report both main effects (averaged over the other factor) and cell means.

- Partial factorial ('unbalanced') designs drop problematic cells – useful when certain combinations are unethical or impractical.
- For > 3 factors, fractional factorial designs (Taguchi) reduce cell count at the cost of confounding higher-order interactions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale